Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering and Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: Basic of NLP**	
Experiment No:- 6	**Date:-**	**Enrollment No:-**92301733001

Code:

Basics of NLP

```
[14] text="""Nishith Kotak is teaching NLP programming in the
class of Semester 6 for the subject AI to
BTech ICT students in MA112, Marwadi University,
Rajkot, India"""

[15] type(text)
     str

[16] text[5]
     't'

[ ] #NLP: SPACY, NLTK, Gensim

[17] import spacy

[18] #load the language model
     nlp=spacy.load('en_core_web_sm')

[19] doc=nlp(text)

[20] #Named Entity Recognition: NER
     for token in doc.ents:
       print(token.text, token.label_)
```

Output:

```
Nishith Kotak PERSON
NLP ORG
BTech ORG
MA112 PRODUCT
Marwadi University ORG
Rajkot GPE
India GPE
```

Code:

```
[21] #POS Tags: Part of Speech tagging
     text="Let's google it"
     doc=nlp(text)
     for token in doc:
       print(token.text,"-->",token.pos_)

     Let --> VERB
     's --> PRON
     google --> VERB
     it --> PRON

[22] #Stemming vs lemmatization
     text="""I studied AI subject and
     then meeting Mr. Sachin in a meeting"""

[23] doc=nlp(text)

[24] doc
     I studied AI subject and
     then meeting Mr. Sachin in a meeting

[25] #stemming
     import nltk
     from nltk.stem.porter import *

[26] p_stem=PorterStemmer()
     for word in text.split():
       print(word,"-->",p_stem.stem(word))

     I --> i
```

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Output:

```
print(word,"-->",p_stem.stem(word))

I --> i
studied --> studi
AI --> ai
subject --> subject
and --> and
then --> then
meeting --> meet
Mr. --> mr.
Sachin --> sachin
in --> in
a --> a
meeting --> meet
```

```
[27]
#lemmatization
for token in nlp(text):
  print(token.text,"-> ",token.pos_,
        "->",token.lemma_)

I ->  PRON -> I
studied ->  VERB -> study
AI ->  PROPN -> AI
subject ->  NOUN -> subject
and ->  CCONJ -> and

 ->  SPACE ->

then ->  ADV -> then
meeting ->  VERB -> meet
Mr. ->  PROPN -> Mr.
Sachin ->  PROPN -> Sachin
in ->  ADP -> in
a ->  DET -> a
meeting ->  NOUN -> meeting
```